Mohammad Sami Nur Islam

GitHub • Google Scholar • LinkedIn • Website • mohammad.sami.islam@mail.mcgill.ca • +1(438)5306811

Education

Mila/McGill University, *Computer Science and Mathematics* Sep 2021 – present | Montreal, Canada

- **MSc (Thesis) Computer Science** under the supervision of Professor Doina Precup [Jan 2025 - April 2027 (expected)]
- **BA Computer Science and Mathematics** (GPA: 3.83) [Sep 2021 - Dec 2024]

Publications

Code as Reward: Empowering Reinforcement Learning with VLMs
David Venuto, **Mohammad Sami Nur Islam**, Martin Klissarov, Doina Precup, Sherry Yang, Ankit Anand [ICML 2024, Spotlight, 3.5% acceptance rate)]

Langevin Soft Actor-Critic: Efficient Exploration through Uncertainty-Driven Critic Learning
Haque Ishfaq, Guangyuan Wang, **Mohammad Sami Nur Islam**, Doina Precup [ICLR 2025]

Scavenging Hyena: Distilling Transformers into Long Convolution Models
Tokiniaina Raharison Ralambomihanta, Shahrad Mohammadzadeh, **Mohammad Sami Nur Islam**, Wassim Jabbour, Laurence Liang [Efficient Systems for Foundation Models (ES-FoMO) workshop @ ICML 2024]

Experiences

Researcher, *MILA - Quebec Artificial Intelligence Institute* Sep 2023 – present | Montreal, Canada

- Investigating offline-to-online RL when offline data is not retained, using expressive frozen actor priors to prevent representation collapse and policy forgetting during fine-tuning (under review at ICLR 2027).
- Scaling unsupervised, goal-conditioned RL to complex control tasks without engineering dense reward functions or sub-tasks.
- Previously, designed reward functions for sparse reward environments using VLMs and applied Thompson Sampling to enhance exploration in deep continuous control.

Machine Learning Research Intern, *RBC Borealis* Sep 2025 – Dec 2025 | Montreal

- Proposed a Conditional LoRA approach for LLM-based recommendation that dynamically modulates representations using Collaborative Filtering and semantic embeddings.
- Evaluated the architecture on Beauty, LastFM, and ML1M datasets, demonstrating competitive performance in standard recommendation benchmarks.

Awards

Master's Research Scholarship, *Fonds de recherche du Québec – Nature et technologies* (*FRQNT*) [$ 33,334] May 2025

Hugh Brock Undergraduate Scholarship, *McGill* [$ 20,000] Sep 2021

Projects

ProjectX Undergraduate Research Competition (by U of T AI), *First-prize* ($10000) Sep 2023 – Jan 2024

- Core member of the **McGill team** investigating cross-architecture knowledge transfer feasibility from slower attention-based **LLM** architectures to the faster, long convolution-based ones like Hyena.
- The research aims to **eliminate** the need to **train** these more efficient models **from scratch**, saving resources.
- Coded the distillation pipeline in **PyTorch**, designed experiments, and distributed tasks.

Skills

Deep Learning / Machine Learning Libraries — PyTorch, Jax, Scikit-learn, NumPy, Pandas |

Programming Languages — Python, SQL, C, Java | **Cloud/MLOps —** SLURM, AWS, Docker, GitHub Actions

Extracurriculars

Bootcamp Technical Project Manager, *McGill AI Society* Apr 2023 – Jun 2024 | Montreal, Canada
Helped redesign the curriculum and the assignments for the boot camp. Mentoring boot campers on their capstone projects.